# pd-trigger-lite-dip — Design Document

## PCB-0016-B

hwde v1 pipeline

generated 2026-09-25 18:10:00

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# 1 Board and Run Metadata

| Field | Value |
|---|---|
| board | pd-trigger-lite-dip |
| part number | PCB-0016-B |
| workspace | `boards/PCB-0016-B_pd-trigger-lite-dip` |
| phase | P9 |
| gate status | all 5 recorded gates pass |
| state created | 2026-09-24T10:16:45 |
| state updated | 2026-09-24T10:44:10 |

# 2 Overview

### Brief: pd-trigger-lite

Owner's words: "a simple pd trigger board like the showcase one ... No need to copy the existing ... Ideally low cost ... the zips and component placement files so I can order PCB cheaply from JLC".

Design intent (set by the dispatching planning session; unattended run - the owner is away, every recipe question is decided here and recorded with state.py decision):

- USB-C PD SINK trigger, 2-layer, 1 oz, small (aim <= ~25 x 15 mm; bigger only if DRC needs it).
- CH224K (LCSC C970725) with its minimal application circuit.

- Voltage selectable after assembly with no tool beyond a soldering iron (solder jumpers / 0R links), sensible assembled default 12 V.
- Output: VBUS + GND on two large 2.54 mm-compatible through-hole pads, no screw terminal. Rated 3 A (no e-marker, so 3 A is the cable/source ceiling).
- One power LED if it costs one basic resistor + one basic LED.
- A bulk/input cap per datasheet. No TVS/fuse unless the datasheet requires it.
- JLC BASIC parts everywhere possible; USB-C connector and CH224K are extended - cheapest widely stocked power-only receptacle exposing CC1/CC2.
- Deliverables: gerber zip, JLC BOM.csv + CPL.csv, order quote for qty 5 with assembly, owner guide PDF, README. Never place an order.
- Showcase boards/pd-trigger: learn from it, reuse its CH224K extraction, do not copy its layout or feature set.

**Variant brief: pd-trigger-lite-dip (2026-09-24)**

Planning session: copy pd-trigger-lite, replace the voltage-select 0603 links with an SMD DIP switch so the output voltage is chosen by switch, with 5 V as one choice. CH224A, 5-position switch one-hot: pos1 CFG1 -> 100k -> VHV (5 V), pos2-5 CFG1 -> 6.8k/24k/56k/120k -> GND. Grow the board only as needed. Same deliverables as lite. No order, no spend.

# 3 Requirements

## Requirements: pd-trigger-lite-dip

Variant of pd-trigger-lite: the solder links are replaced by a 5-position DIP switch (5 V added).

Source: `brief/brief.md`. Written inline by the orchestrator (no requirements-analyst spawn: a one-block board, recorded as a decision). Unattended run: every open item below was answered by the orchestrator as the owner's delegate and recorded with `state.py decision`. Mode: none (the brief carries no mode token) - design normally.

### 1. Function

A low-cost USB-C Power Delivery trigger. A PD sink controller negotiates a fixed voltage from a USB-C PD charger and passes VBUS straight through to two output pads. No regulation, no conversion, no switching element in the power path. The requested voltage is chosen by a 5-position DIP switch: 5, 9, 12, 15 or 20 V.

### 2. Interfaces

- Input: USB-C receptacle, power-only use, PD SINK. CC1/CC2 to the controller; D+/D- to the controller's DP/DM per the datasheet reference (no USB data function).
- Output: VBUS and GND on two 2.54 mm-pitch plated through-holes (take a 1x2 header or wires). No screw terminal (cost).
- Voltage selection: SW1, one switch per voltage (5/9/12/15/20 V), exactly one ON, set before plugging in. All assembled at JLC; the switch is an extended part.

**3. Power**

- Input: USB-C PD source, 5 V default, negotiated 9/12/15/20 V.
- Output current rating: 3 A continuous at any voltage (a non-e-marked cable caps the contract at 3 A; the board makes no 5 A claim).
- Controller supply: from VBUS directly (CH224A VHV pin is rated 32 V).

**4. Environment**

Bench use, indoor, 0-50 C ambient. No enclosure, no battery, no mains.

**5. Size & mounting**

Target <= 25 x 15 mm (soft; grow only if DRC needs it). No mounting holes (cost/size; the board hangs off its cable and output wires).

**6. Quantity & budget**

Quantity 5, JLCPCB PCB + economy PCBA, lowest cost. JLC Basic parts wherever possible.

**7. Assembly**

Single-sided SMT at JLC. The 1x2 output header is optional and hand-fitted by the owner.

**8. Compliance/safety flags**

- >3 A: NO (3 A max). Mains: NO. Battery: NO.
- Protection: no TVS/fuse - the controller datasheet requires none (decision recorded).

**9. Open questions**

None open. Every question was answered by the orchestrator as delegate (see state.json decisions): controller part, selection method, default voltage, output format, current rating, LED, bulk cap, protection.

# 4 Architecture

## Decisions of record (state.json)

| Phase | What | Why |
|---|---|---|
| P0 | D2 voltage select = SW1 5-position DIP, one-hot: pos1 100k to VHV = 5 V, pos2-5 6.8k/24k/56k/120k to GND = 9/12/15/20 V | Owner asked for switch selection with 5 V; CH224 manual 5.2.1, 5.2.2, 6.1.2 |
| P0 | D8 SW1 = SHOU HAN 1.27-5P TPPT C7421518 (extended, 0.53 USD) | Most-stocked 5P 1.27 mm SMD DIP JLC can place; 24 V rating covers the 20 V open-contact case |
| P0 | D9 board 25 x 21 mm | Switch block needs 6 mm more height than lite's link grid; rest of lite layout moved down |
| P0 | D10 silk: voltage per switch row plus <ON mark; R1-R5 refdes hidden | ON side assumed from DIP convention (datasheet silent); guide defers to the ON printed on the switch |

### Sheet plan

#### pd-trigger-lite - sheet plan

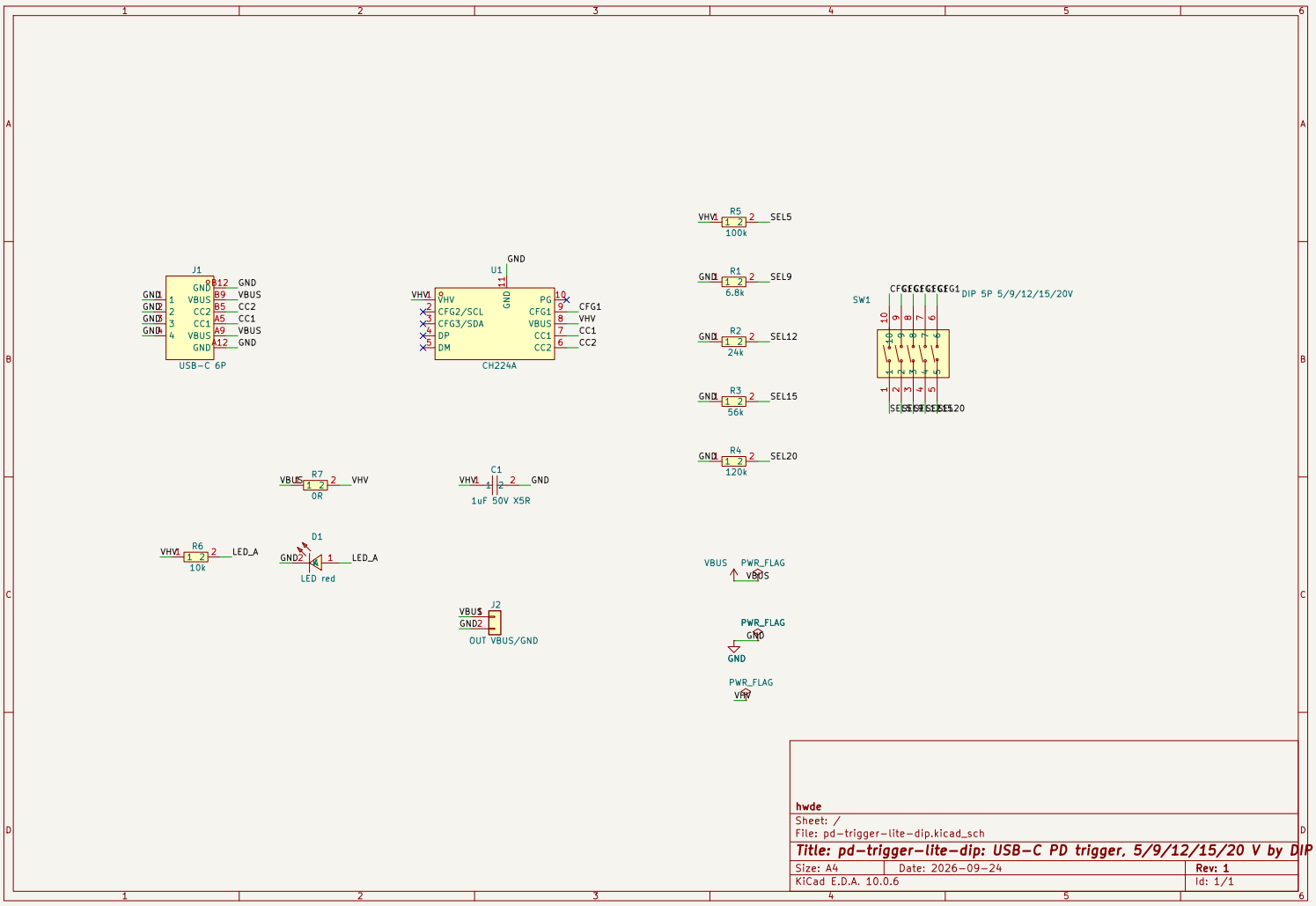
One flat root sheet `kicad/pd-trigger-lite.kicad_sch`, pwr_base 1, generator `kicad/gen/root.py`. 14 symbols, one chain, every cross-block net is a rail (VBUS, GND) or a two-part stub - hierarchy would buy nothing.

Canonical nets: `VBUS`, `GND` (power symbols, bare); `/CC1`, `/CC2`, `/CFG1`, `/SEL5`, `/SEL9`, `/SEL12`, `/SEL15`, `/SEL20` (resistor-to-switch nodes), `/LED_A`.

Full architecture narrative (not inlined; see the artifact index): `architecture/blocks.md`, `architecture/decisions.md`.

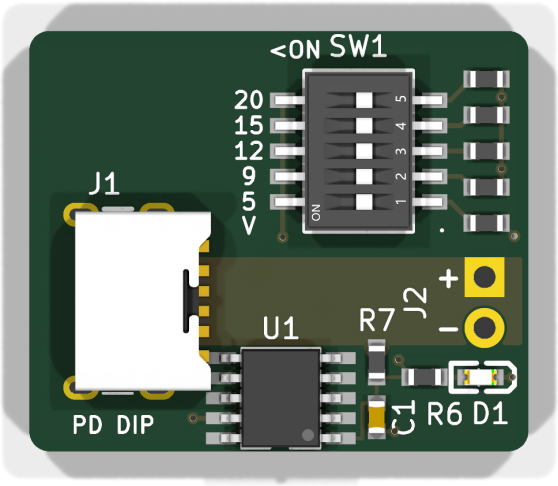
## 5 Schematic

Gate `erc` (phase P4): **pass** after 2 attempt(s); last 2026-09-24T10:21:38, failing 0 of 0. The full schematic PDF follows.

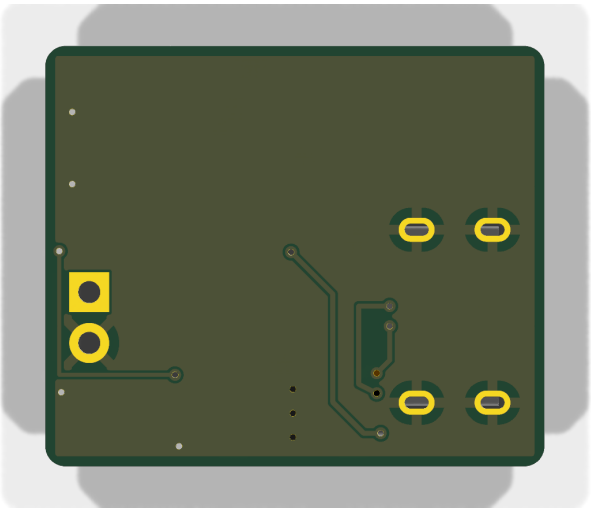

J1
USB-C 6P
U1
CH224A
R5
100k
R1
6.8k
R2
24k
R3
56k
R4
120k
SW1
DIP 5P 5/9/12/15/20V
R7
0R
C1
1uF 50V X5R
R6
10k
D1
LED red
J2
OUT VBUS/GND
PWR_FLAG
hwde
Sheet: /
File: pd-trigger-lite-dip.kicad_sch
Title: pd-trigger-lite-dip: USB-C PD trigger, 5/9/12/15/20 V by DIP switc
Size: A4
Date: 2026-09-24
Rev: 1
KiCad E.D.A. 10.0.6
Id: 1/1

# 6 Layout

Gate `place` (phase P6): **pass** after 1 attempt(s); last 2026-09-24T10:33:52, failing 0 of 0.
Gate `drc_routed` (phase P7): **pass** after 1 attempt(s); last 2026-09-24T10:33:54, failing 0 of 0.




`reports/pd-trigger-lite-dip_top.png`



`reports/pd-trigger-lite-dip_bottom.png`

# 7 Verification

Gate `verify` (phase P8): **pass** after 1 attempt(s); last 2026-09-24T10:33:58, failing 0 of 11. *reports/verify_all.json not available.*

# 8 DFM and Fabrication

Gate `dfm` (phase P9): **pass** after 1 attempt(s); last 2026-09-24T10:35:15, failing 0 of 0. DFM findings by severity: none.

### Fabrication outputs

Exported layers: F.Cu, B.Cu, F.Silkscreen, B.Silkscreen, F.Mask, B.Mask, F.Paste, B.Paste, Edge.Cuts.

### Bill of materials

| Comment | Designator | Footprint | LCSC |
|---|---|---|---|
| 1uF 50V X5R | C1 | C0603 | C15849 |
| LED red | D1 | LED-SMD_L1.6-W0.8-R-RD | C2286 |
| USB-C 6P | J1 | USB-C-SMD_918-418K2023S40006 | C2798175 |
| 6.8k | R1 | R0603 | C23212 |
| 24k | R2 | R0603 | C23352 |
| 56k | R3 | R0603 | C23206 |
| 120k | R4 | R0603 | C25808 |
| 100k | R5 | R0603 | C25803 |
| 10k | R6 | R0603 | C25804 |
| 0R | R7 | R0603 | C21189 |
| DIP 5P 5/9/12/15/20V | SW1 | SW-SMD_5P-P1.27_1.27-5P | C7421518 |
| CH224A | U1 | ESSOP-10_L4.9-W3.9-P1.0-LS6.0-TL-EP | C42459160 |

0 CPL rotation correction(s) applied; no missing LCSC numbers. *fab/order.json not available.*

## 9 Run Record (Appendix)

### Phase digests

**P0** *(no digest recorded)*

**P1** *(no digest recorded)*

**P2** *(no digest recorded)*

**P3** *(no digest recorded)*

**P4** *(no digest recorded)*

**P5** *(no digest recorded)*

**P6** *(no digest recorded)*

**P7** *(no digest recorded)*

**P8** *(no digest recorded)*

**P9** *(no digest recorded)*

### Gate attempt history

| Timestamp | Gate | Status | Failing/Total |
|---|---|---|---|
| 2026-09-24T10:20:42 | erc | fail | 11/11 |
| 2026-09-24T10:21:38 | erc | pass | 0/0 |
| 2026-09-24T10:33:52 | place | pass | 0/0 |
| 2026-09-24T10:33:54 | drc_routed | pass | 0/0 |
| 2026-09-24T10:33:58 | verify | pass | 0/11 |

| Timestamp | Gate | Status | Failing/Total |
|---|---|---|---|
| 2026-09-24T10:35:15 | dfm | pass | 0/0 |

### Issues

*(no entries)*

### Human checkpoints

| Id | Phase | Status | When | Note |
|---|---|---|---|---|
| 1 | P2 | not reached | | |
| 2 | P4 | not reached | | |
| 3 | P6 | not reached | | |
| 4 | P8 | not reached | | |
| 5 | P10 | not reached | | |

### Generation cost

Model cost of generating this board: **$6.31**.

| Step | Cost |
|---|---|
| Before the workspace (brief and skill read) | $1.13 |
| Schematic | $0.85 |
| Placement | $2.10 |
| DFM, order files and guide | $2.23 |

Not in the total: pre-buy list and BOM fixes, shared with the hwde skill's pre-buy list feature, cost $4.93.

Guide rebuilds by later skill-wide rounds (board renders with parts, fresh guide renders, this cost record) are skill work and are not counted; the fresh-guide-renders round left no transcript on this host.

The review-fix round resumed the design session, so its cost is inside the design run; the loop log shows only the first iteration.

The worker loop logged $6.15.

## 10 Artifact Index

| File | Size | Note |
|---|---|---|
| `state.json` | 11,864 B | |
| `kicad/PCB-0016-B_pd-trigger-lite-dip.kicad_pcb` | 88,278 B | |
| `kicad/PCB-0016-B_pd-trigger-lite-dip.kicad_sch` | 80,953 B | |
| `fab/PCB-0016-B_pd-trigger-lite-dip_gerbers.zip` | 17,266 B | |
| `fab/BOM.csv` | 415 B | |
| `fab/CPL.csv` | 431 B | |
| `brief/brief.md` | 1,793 B | |
| `requirements.md` | 2,429 B | |
| `architecture/blocks.md` | 1,896 B | |
| `architecture/constraints.json` | 1,276 B | |
| `architecture/decisions.md` | 2,126 B | |
| `architecture/sheets.md` | 411 B | |
| `reports/board_init.json` | 10,716 B | |

| File | Size | Note |
|---|---|---|
| reports/bom_cpl.json | 9,656 B | |
| reports/cost.json | 4,526 B | |
| reports/drc_final.json | 588 B | |
| reports/fab_export.json | 2,370 B | |
| reports/gate-dfm.json | 1,563 B | |
| reports/gate-drc_routed.json | 1,072 B | |
| reports/gate-erc.json | 873 B | |
| reports/gate-place.json | 1,457 B | |
| reports/gate-verify.json | 2,065 B | |
| reports/guide_facts.json | 8,200 B | |
| reports/lib_pull.json | 3,794 B | |
| reports/pd-trigger-lite-dip_bottom.png | 54,836 B | |
| reports/pd-trigger-lite-dip_top.png | 218,940 B | |
| reports/place_edit_p6.json | 1,625 B | |
| reports/place_edit_silk.json | 1,933 B | |
| reports/place_p6.ops.json | 1,429 B | |
| reports/planes_gen.json | 1,173 B | |
| reports/render.json | 470 B | |
| reports/route_edit_manual.json | 8,995 B | |
| reports/route_edit_manual_result.json | 6,055 B | |
| reports/rules_gen.json | 1,044 B | |
| reports/schematic.pdf | 84,556 B | |
| reports/silk_ops_manual.json | 2,064 B | |
| reports/silk_place.json | 3,411 B | |
| reports/silk_place_ops.json | 1,443 B | |